Promotions & Pricing Analytics – Business Case

This project was developed as part of a technical assessment using a fictional business scenario.

BUSINESS OBJECTIVE

Analyze the product and promotional base of a fictional retail company

Data

Identify patterns in revenue, discounts, and promotions.

Analysis

Supporting the Quality Management team with actionable insights.

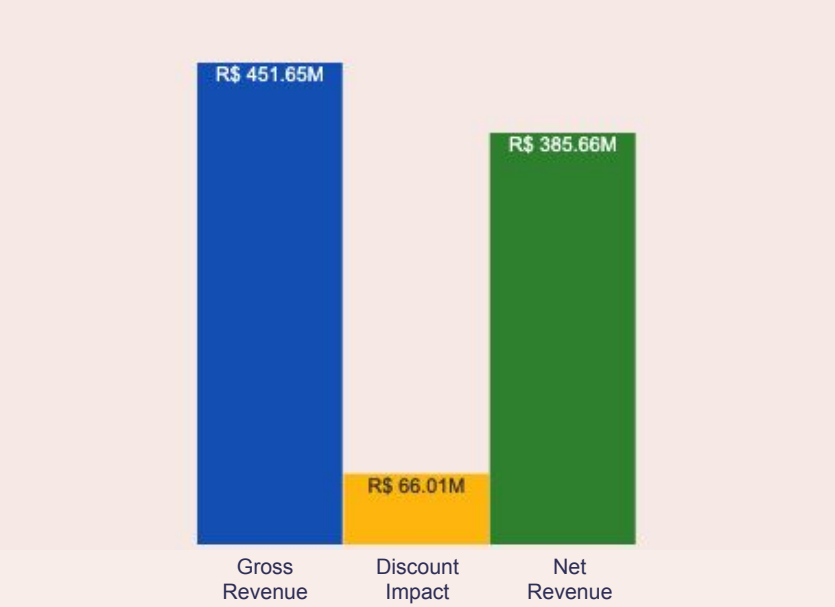
Decision

HOW I PREPARED THE DATA

- Information: 19,502 records and 20 columns
- Duplicate removal: 200 rows → final dataset: 19,302 records
- Handling of missing values:
  - vlr_desconto_real and vlr_desconto_real2: filled with median
  - cod_ano: filled based on the first 4 digits of cod_ciclo
- Types: cod_agrupador_sap_material → converted from float64 to string
- Categorical columns with dominance (90%): None
- Outliers:
  - Keep, as they may represent aggressive promotions or legitimate pricing strategies

IMPACT OF DISCOUNTS ON REVENUE

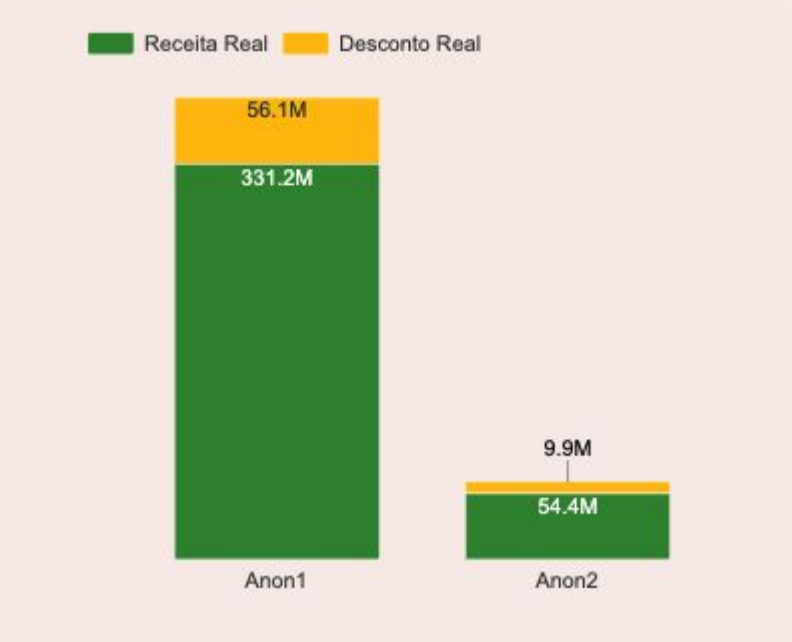
An analysis of the fictional retail company's revenue reveals a significant impact of discounts on sales, especially given the high share of products sold under promotional pricing.



84%

of products rely on discounts
→ direct risk of margin erosion.

PRODUCTS CONCENTRATING REVENUE AND DISCOUNTS



The top product accounts for

85.9% of net revenue,

but is subject to an average discount of

14.5%, creating significant

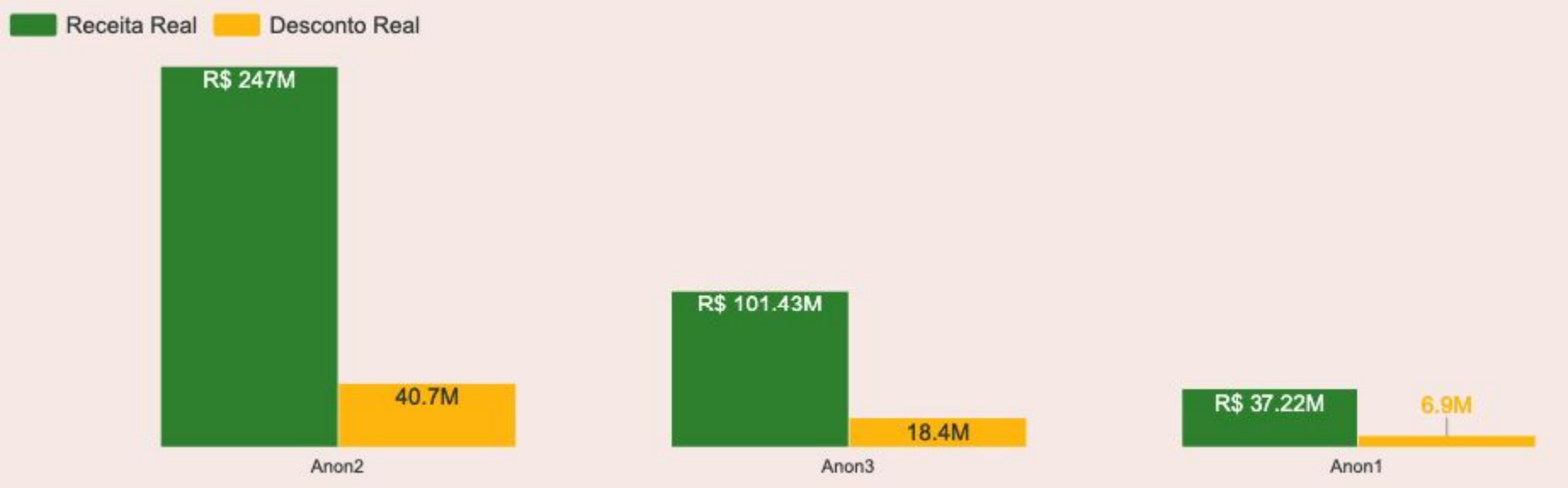
pressure on portfolio profitability.

KEY DIFFERENCES BETWEEN BRANDS

Marca do Produto	Receita Real ▾	Receita Tabelada	Desconto Médio
Anon5	R$ 212.15M	R$ 250.41M	15.28%
Anon1	R$ 89.37M	R$ 104.1M	14.15%
Anon8	R$ 36.56M	R$ 44.23M	17.34%
Anon2	R$ 35.13M	R$ 39.11M	10.2%
Anon4	R$ 5.27M	R$ 5.8M	9.2%
Anon3	R$ 3.7M	R$ 4.07M	9.05%
Anon6	R$ 2.75M	R$ 3.07M	10.47%
Anon7	R$ 726.32K	R$ 841.71K	13.71%
Anon9	R$ 63.51	R$ 63.51	0%

Brands Anon5 and Anon8 have **discounts above 15%**; however, Anon8 shows lower sales volume, suggesting **promotions without proportional returns**.

REVENUE AND DISCOUNTS BY STATE



Anon2 is the largest market in terms of revenue and discount usage → a potential risk: are discounts **driving sales or simply eroding margins**?

DISCOUNT STRATEGIES BY CHANNEL AND CUSTOMER SEGMENT

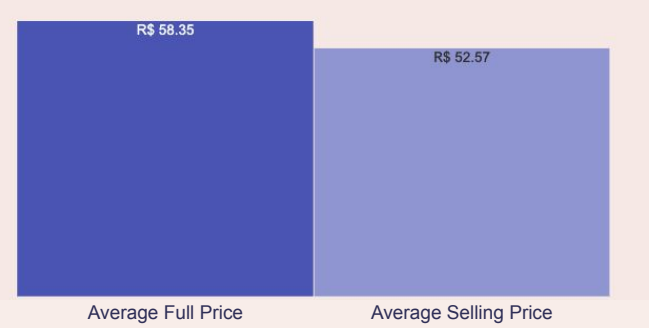
Canal	Público	Receita Real	Desconto Total
Anon1	Anon2	75.2M	1.4M
	Anon1	65.9M	20.6M
	Anon3	43.5M	18.1M
	Anon4	9.9M	2.9M
	Anon5	20.6K	71
	Total	**194.5M**	**43M**
Anon2	Anon1	98M	21.8M
	Anon2	72.3M	190....
	Anon3	20.8M	1.1M
	Anon4	1.8K	18.5
	Total	**191.1M**	**23M**

Customer segment Anon1 concentrates R$ 42.4M in discounts — significantly higher than others — putting pressure on margins across both channels.

IMPACT OF PROMOTIONS AND PRICING

Net Revenue by Promotion Type (Reseller)

Dependence on a single promotion mechanism (R$ 311M in reseller sales) limits flexibility and reduces profitability.



Products are sold with an average discount of 9,9% → nearly the entire portfolio is priced below full price.

STRATEGIC RECOMMENDATIONS

Implement active discount management to reduce margin erosion

Adopt segmented pricing by category, brand, and customer segment

Diversify promotion mechanisms for resellers